Research Statement

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Cure rate models have gained attention as traditional survival analysis often neglects individuals who remain event-free after long follow-up periods. These cured or insusceptible subjects are explicitly accounted for in cure models, which estimate both the cured proportion and the survival distribution for the susceptible group.

My research addresses methodological challenges in short- and long-term survival modeling, where heterogeneity in cure propensity and latency dynamics cannot be fully captured by conventional models. I develop flexible cure rate models that relax restrictive assumptions in both the incidence (cure probability) and latency (time-to-event) components. The introduced flexibility enables the modeling of asymmetric, dependent, and nonlinear relationships between covariates and survival outcomes. In parallel, I incorporate structured variable selection to identify key predictors and capture shared covariate structures across components—an essential feature in high-dimensional biomedical settings, such as genetic or proteomic studies. Collectively, these developments aim to create more adaptive and interpretable frameworks for understanding complex survival mechanisms. A brief summary of my current projects and future research directions is provided below.

Flexible mixture cure rate model

The mixture cure rate model (MCM) is a special type of survival model that divides the study population into cured and uncured subgroups (Berkson & Gage, 1952). It models the overall survival as a mixture of two components: the incidence, representing the probability of being uncured, and the latency, representing the survival function of the susceptible individuals. Conventionally, the incidence component is modeled using logistic regression with a symmetric logit link function. The limitation of this standard approach is critical in scenarios with extreme cure rates: the logit function imposes an assumption of equal influence on both outcomes, which fails to capture the true heterogeneity and imbalance present, potentially leading to inaccurate parameter inference (Czado & Santner, 1992).

To address this fundamental deficiency, we introduce a novel, **flexible MCM based on asymmetric link functions**. Out model uses the power and reciprocal power transformations of the logit (Bazán *et al.*, 2017; Nagler, 1994), effectively generalizing the incidence component. The key advantage of this asymmetric is its ability to finely tune the model's response to different degrees of imbalance. This results in a highly adaptable and interpretable model that can more precisely estimate cure probabilities, particularly in datasets where the proportion of cured individuals is near zero or near one.

To support this flexible model, we further designed two complementary and robust estimation frameworks. First, a fully Bayesian implementation, developed in `Stan`, enables robust parameter estimation with coherent uncertainty quantification and the ability to incorporate prior knowledge for stable inference. In parallel, we developed an efficient Expectation-Maximization (EM Dempster *et al.* (1977)) algorithm, which achieves faster convergence and scalabiltiy to large or high-dimensional datasets. To promote reproducibility and accessibility, we implemented the high=performance EM-based framework into the open-source `R` package, `EMGCR`. Simulation studies demonstrate the framework's strong performance even under extreme imbalance (e.g., cure fractions up to 70% and 45% censoring among the uncured), showing its superior stability compared to symmetric models. Applications to criminology recidivism and liver cancer datasets further highlight its unprecedented robustness and practical utility, making it a definitive advancement in cure rate modeling. Crucially, the architecture of the `EMGCR` package is deliberately designed for possible extension, readily allowing the incorporation of other flexible link functions and different choices of survival distributions for future model improvements, making it a foundation for addressing even more challenging cure rate scenarios.

Bayesian Modeling of Structured Sparsity in Mixture Cure Models

The traditional Mixture Cure Model (MCM) assumes independence between the incidence (cure probability) and latency (survival time of the uncured) components, treating their respective regression coefficients (β_j, η_j) as unrelated. This assumption fails when biological or clinical factors, such as tumor markers or cancer staging, exert a shared, structured influence on both the probability of cure and the subsequent survival rate. Ignoring this relationship leads to an inferential flaw: it risks misidentifying significant covariates, weakening statistical power, and producing less interpretable results when seeking joint biomarkers (Fu *et al.*, 2022; Shi *et al.*, 2020).

To address this, we developed a **correlated spike-and-slab (CSS) prior** for Bayesian variable selection within the MCM. The key contribution lies in moving beyond independent priors to model the joint distribution of (β_j, η_j) for each shared covariate j. The CSS prior is formulated as a four-component bivariate Gaussian mixture distribution, where each component directly represents a distince pattern of covariate effect:

$$(\beta_j, \eta_j)^\top | \{V_k\}_{k=1}^3 \sim \sum_{k=1}^{4} \pi_k \mathsf{N}_2(\mathbf{0}, \mathbf{V}_k) + \pi_2 \mathsf{N}_2(\mathbf{0}, \mathbf{V}_2) + \pi_3 \mathsf{N}_2(\mathbf{0}, \mathbf{V}_3) + \pi_4 \mathsf{N}_2(\mathbf{0}, \mathbf{V}_4). \tag{1}$$

- $\mathbf{V}_1$ (Dual-Effect, π_1): This component includes covariance matrix, which allows for non-zero correlation between β_j and η_j. This structure promotes joint selection of covariates and enforces sign consistency when a single marker influences both cure and survival in a related manner (e.g., better prognosis implies both higher cure chance and longer latency).
- $\mathbf{V}_2/\mathbf{V}_3$ (Incidence-Only/Latency-Only, π_2, π_3): These account for covariates that impact only one component.
- $\mathbf{V}_4$ (No-Effect, π_4): The standard spike component indicating irrelevance.

Based on the posterior samples obtained, we identify important covariates using the posterior inclusion probabilities (PIPs). We classify for each covariate to the four effect categories, and the marginal probabilities of inclusions, $\Pr(\beta_j \neq 0|D)$ and $\Pr(\eta_j \neq 0|D)$, are then obtained by summing the probabilities of the corresponding patterns.

This work significantly advances high-dimensional survival modeling by providing the first formal Bayesian framework that unifies structured sparsity and cure rate analysis. Applications to the TCPA breast cancer dataset successfully identified key protein markers, especially the biomarkers that were jointly influential on cure probability and survival latency patterns that were undetectable under independent selection. The CSS-MCM framework is a generalizable solution, where it can be extended to other structured sparsity problems in complex survival models, particularly in the context of high-dimensional genomics and precision oncology, where identifying correlated multi-target biomarkers is essential.

Joint Gaussian Process incorporated promotion time cure model

The Promotion-Time Cure Model (PCM) provides a broader framework than the Mixture Cure Model (MCM) by replacing the binary cure split with a count distribution that directly relates the number of underlying risk factors to the incidence component (Yakovlev & Tsodikov, 1996; Chen *et al.*, 1999). While previous work has incorporated a standard Gaussian Process (GP) to capture non-linear covariate effects within the PCM's cure fraction (Li *et al.*, 2016), the model is still unable to coherently capture non-linear effects that simultaneously and correlatively influence both the cure and latency components.

Our current research addresses this limitation by introducing a **joint GP layer within the promotion-time cure framework (GP-PCM)**. To flexibly capture complex, non-linear covariate effects z_i that jointly influence both components, we impose a single latent GP $f = (f_1, \dots, f_N)^\top \sim \mathcal{N}(0, K_\theta)$ on both the cure and latency linear predictors:

$$(\lambda_{\mu i}, \theta_{hi}) = (\mathbf{x}_{\mu i}^\top \boldsymbol{\beta}_\mu + f_i,\ \mathbf{x}_{hi}^\top \boldsymbol{\beta}_h + \rho f_i)$$

The parameter $\rho \in [-1, 1]$ acts as a correlation parameter governing the dependence of the non-linear effects between the incidence and latency components. When $\rho = 0$, the non-linear effect is restricted to the cure component, recovering the approach of Li *et al.* (2016). When $|\rho|$ is non-zero, the model allows the non-linear impact of z_i to borrow information across submodels, facilitating a coherent, structured non-parametric adjustment for continuous covariates.
The methodological framework for GP-PCM has been fully developed. We are currently engaged in extensive simulation studies to rigorously assess the model's performance under various correlation and non-linearity regimes, specifically focusing on the inferential properties of the correlation parameter ρ. The goal for this simulation is to demonstrate that the information-borrowing structure governed by ρ achieves reduced estimation bias and improved calibration compared to models that treat non-linear effects independently.

Future research directions

Building on my current work, I envision my future research to expand the scope of survival analysis and statistical genomics in the following directions but not limited to:

Integration of causal inference. Despite the broad development of causal inference methodology in survival analysis, relatively few approaches have been proposed for causal inference under cure rate models. Because treatment may influence both the probability of being cured and the latency distribution among the uncured, careful methodological development is needed to disentangle and estimate these distinct causal components. Recent work by Wang *et al.* (2024) represents an important step by defining four principal strata according to potential cure status and jointly estimating treatment effects on incidence and latency. Building on this framework, future research could introduce correlated treatment-effect structures within the cure–latency decomposition, allowing dependencies between incidence and latency effects to be explicitly modeled. Such extensions would provide more realistic causal characterizations of treatment mechanisms while improving efficiency and interpretability in settings where cure is a meaningful feature of the event process.

High-dimensional multi-omics integration A promising objective of my future research is to extend my work on structured sparsity to the integration of bulk multi-omics data. To address the challenges, I plan to develop hierarchical variable selection frameworks that incorporate the correlated spike-and-slab priors across different data. This approach will allow for the identification of shared biomarkers that drive the traits for specific cells, while preserving the underlying biological signals of the distinct cell types. Furthermore, I intend to work further to leverage the Gaussian process prior structures for spatial-temporal modeling. By combining structured variable selection with non-parametric modeling, this framework will provide a flexible approach for capturing complex key drivers.

Bibliography

Bazán, J., Torres-Avilés, F., Suzuki, A. & Louzada, F. (2017). Power and reversal power links for binary regressions: An application for motor insurance policyholders. *Applied Stochastic Models in Business and Industry*, 33, 22–34.

Berkson, J. & Gage, R. P. (1952). Survival curve for cancer patients following treatment. *Journal of the American Statistical Association*, 47(259), 501–515.

Chen, M. H., Ibrahim, J. G. & Sinha, D. (1999). A new bayesian model for survival data with a surviving fraction. *Journal of the American Statistical Association*, 94(447), 909–919.

Czado, C. & Santner, T. J. (1992). The effect of link misspecification on binary regression inference. *Journal of Statistical Planning and Inference*, 33(2), 213–231.

Dempster, A. P., Laird, N. M. & Rubin, D. B. (1977). Maximum likelihood from incomplete data via the EM algorithm. *Journal of the Royal Statistical Society. Series B (Methodological)*, 39(1), 1–38.

Fu, H., Nicolet, D., Mrózek, K., Stone, R. M., Eisfeld, A. K., Byrd, J. C. & Archer, K. J. (2022). Controlled variable selection in weibull mixture cure models for high-dimensional data. *Statistics in Medicine*, 41(22), 4340–4366. Epub 2022 Jul 6.

Li, D., Wang, X. & Dey, D. K. (2016). A flexible cure rate model for spatially correlated survival data based on generalized extreme value distribution and gaussian process priors. *Biometrical Journal*, 58(5), 1178–1197.

Nagler, J. (1994). Scobit: An alternative estimator to logit and probit. *American Journal of Political Science*, 38(1), 230–255.

Shi, X., Ma, S. & Huang, Y. (2020). Promoting sign consistency in the cure model estimation and selection. *Statistical Methods in Medical Research*, 29(1), 15–28.

Wang, Y., Deng, Y. & Zhou, X.-H. (2024). Causal inference for time-to-event data with a cured subpopulation. *Biometrics*, 80(2), ujae028.

Yakovlev, A. Y. & Tsodikov, A. D. (1996). *Stochastic models of tumor latency and their biostatistical application*. World Scientific, Singapore.